

Lecture 15, part 1: Predicting Out-of-Distribution Performance

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Instructors: A. Vijayaraghavan and V. Srinivas

Scribe: Baenan McKeown

We saw a presentation on *Leveraging Unlabeled Data to Predict Out-of-Distribution Performance* [Garg, Balakrishnan, Lipton, Neyshabur, and Sedghi, 2022], given by Nathan Reimer.

1 ATC: Averaged Thresholded Confidence

1.1 Binary Classification Interpretation

A central challenge in deployment is estimating a model's accuracy on a target distribution Q without access to target labels. The **Averaged Thresholded Confidence (ATC)** method addresses this by using the model's own confidence scores as a proxy for correctness.

Assign a binary correctness label to each source example:

$$z = 1 \quad \text{if the model } f \text{ is correct on } (x, y), \quad z = 0 \text{ otherwise.}$$

ATC acts like a classifier that predicts z from the scalar confidence score $s(f(x))$. A threshold t is selected on the labeled source data so that the number of source examples whose confidence falls below t equals the number of source errors:

$$\#\{i : s(f(x_i)) < t\} = \#\{i : f(x_i) \neq y_i\}.$$

This choice yields the following key counting identity among source points:

$$\begin{aligned} \#(\text{Incorrect}) &= \#(\text{Incorrect Below } t) + \#(\text{Incorrect Above } t), \\ \#(\text{Below } t) &= \#(\text{Incorrect Below } t) + \#(\text{Correct Below } t). \end{aligned}$$

Because $\#(\text{Below } t) = \#(\text{Incorrect})$ by construction,

$$\#(\text{Correct Below } t) = \#(\text{Incorrect Above } t).$$

Intuitively, the correct predictions that exist below the threshold are exactly balanced by the incorrect predictions that exist above it. The fraction of target distribution examples with confidence below t is then used as the ATC estimate of target error rate.

1.2 Calibration

Softmax outputs are not always well calibrated—modern neural networks tend to be *overconfident*, assigning high-confidence scores even to incorrect predictions. A standard remedy is temperature scaling:

$$\hat{q} = \max_j \text{softmax}\left(\frac{s_j}{T}\right),$$

where $T > 1$ is a scalar temperature chosen on a held-out validation set. Dividing the logits by $T > 1$ softens the softmax distribution, reducing overconfidence before computing confidence scores for ATC.

2 Spurious Correlations and OOD Generalization

2.1 Toy Model Setup

To study how spurious correlations affect both classifier behaviour and accuracy estimation, consider the following two-dimensional model:

$$x = (x_{\text{inv}}, x_{\text{sp}}) \in \mathbb{R}^2,$$

where x_{inv} is an invariant feature (genuinely predictive of y across domains) and x_{sp} is a spurious feature defined by

$$x_{\text{sp}} = y \quad \text{with probability } p_{\text{sp}} \in (0.5, 1.0).$$

Thus x_{sp} is correlated with the label in the source distribution, but this correlation may weaken or reverse under distribution shift. The conditional distribution of x_{sp} is supported on $\{-c, c\}$ for some $c > 0$, with $P(x_{\text{sp}} = c \mid y = 1) = p_{\text{sp}}$ and $P(x_{\text{sp}} = -c \mid y = -1) = p_{\text{sp}}$.

2.2 Dependency on the Spurious Feature

Consider classifiers of the softmax form

$$f(x) = \left[\frac{1}{1 + e^{w \cdot x}}, \frac{e^{w \cdot x}}{1 + e^{w \cdot x}} \right].$$

When trained on finite samples, gradient-based optimisation causes the classifier to rely on the spurious feature x_{sp} rather than (or in addition to) the invariant feature x_{inv} .

2.3 Statistical Skew

Let

$$h(x) = w_{\text{inv}} x_{\text{inv}} + w_{\text{sp}} x_{\text{sp}}$$

be a linear classifier. When weights are initialised near 0, the exponential loss $\exp(-y w \cdot x)$ is approximately 1 on all training points. The gradient update for each feature dimension i is proportional to

$$\sum_{(x,y)} y x_i.$$

For the spurious feature, the expected gradient signal is

$$\mathbb{E}[y x_{\text{sp}}] \propto p_{\text{sp}} - (1 - p_{\text{sp}}) = 2p_{\text{sp}} - 1,$$

which is strictly positive for $p_{\text{sp}} > 0.5$. Because the spurious feature provides a stronger (and faster) gradient signal than the invariant feature, gradient descent preferentially increases w_{sp} , causing the learned model to rely on x_{sp} .

2.4 ATC Consistency Under Spurious Correlations

Despite the classifier’s reliance on spurious features, ATC retains its validity provided the spurious weight is non-negative.

Theorem (Informal). Given a classifier in the above setting with $w_{\text{sp}} \geq 0$, ATC using the maximum-confidence score function obtains a *consistent* estimate of the target accuracy.

The formal guarantee is the following. Let n be the number of unlabeled target samples and m be the number of labeled source samples.

Theorem. Let t be selected such that

$$\sum_{i=1}^n \mathbf{1} \left[\max_j f_j(x_i) < t \right] = \sum_{i=1}^m \mathbf{1} \left[\arg \max_j f_j(x_i) \neq y_i \right].$$

Then with probability at least $1 - \delta$,

$$\left| \mathbb{E}_{x \sim P} \left[\mathbf{1} \left[\max_j f_j(x) < t \right] \right] - \mathbb{E}_{(x,y) \sim Q} \left[\mathbf{1} \left[\arg \max_j f_j(x) \neq y \right] \right] \right| \leq \sqrt{\frac{\log(8/\delta)}{n+m}}.$$

The bound decays at the standard parametric rate $O(1/\sqrt{n+m})$, showing that ATC provides a statistically efficient and reliable estimator of target error even when the model exploits spurious correlations, as long as the spurious weights are non-negative.

References

Saurabh Garg, Sivaraman Balakrishnan, Zachary Chase Lipton, Behnam Neyshabur, and Hanie Sedghi. Leveraging unlabeled data to predict out-of-distribution performance. In *ICLR, 2022*. URL <https://arxiv.org/abs/2201.04234>.